
Hydrometry — Open channel flow measurement using thin-plate weirs

*Hydrométrie — Mesure de débit dans les canaux découverts au moyen
de déversoirs à paroi mince*



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Contents

Page

Foreword.....	v
1 Scope	1
2 Normative references	1
3 Terms and definitions.....	1
4 Symbols and abbreviated terms	1
5 Principle.....	2
6 Installation	2
6.1 General.....	2
6.2 Selection of site	2
6.3 Installation conditions.....	2
7 Measurement of head	4
7.1 Head measuring devices.....	4
7.2 Stilling or float well.....	5
7.3 Head-measurement section.....	5
7.4 Head-gauge datum (gauge zero).....	5
8 Maintenance	6
9 Rectangular thin-plate weir.....	6
9.1 Types.....	6
9.2 Specifications for the standard weir.....	8
9.3 Specifications for installation.....	8
9.4 Determination of gauge zero	8
9.5 Discharge formulae — General	11
9.6 Formulae for the basic weir form (all values of b/B)	11
9.7 Formulae for full-width weirs ($b/B = 1,0$)	14
10 Triangular-notch thin-plate weir.....	16
10.1 Specifications for the standard weir.....	16
10.2 Specifications for the installation	18
10.3 Specifications for head measurement.....	18
10.4 Discharge formulae — General	19
10.5 Formula for all notch angles between $\pi/9$ and $5\pi/9$ radians (20° and 100°).....	19
10.6 Formula for specific notch angles (fully-contracted weir)	21
10.7 Accuracy of discharge coefficients — Triangular-notch weirs	22
11 Uncertainties of flow measurement	22
11.1 General.....	22
11.2 Combining measurement uncertainties	23
11.3 Uncertainty of discharge coefficient $u^*(C_d)$ for thin-plate weirs	25
11.4 The uncertainty budget	25
12 Example	25
12.1 General.....	25
12.2 Characteristics — Gauging structure	26
12.3 Characteristics — Gauged head instrumentation.....	26
12.4 Discharge coefficient.....	26
12.5 Discharge estimate	26
12.6 Uncertainty statement	27
Annex A (informative) Flow measurement with small weir tanks	29

Annex B (informative) **Guide to the design and installation of a flow straightener**..... 31

Annex C (informative) **Introduction to measurement uncertainty**..... 33

Annex D (informative) **Sample measurement performance for use in hydrometric worked examples** 41

Annex E (informative) **Specimen tables** 43

Bibliography 59

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Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

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The main task of technical committees is to prepare International Standards. Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights.

ISO 1438 was prepared by Technical Committee ISO/TC 113, *Hydrometry*, Subcommittee SC 2, *Flow measurement structures*.

This second edition cancels and replaces the first edition (ISO 1438-1:1980), of which it constitutes a technical revision. It also incorporates the Amendment ISO 1438-1:1980/Amd 1:1988.

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Hydrometry — Open channel flow measurement using thin-plate weirs

1 Scope

This International Standard defines the requirements for the use of rectangular and triangular (V-notch) thin-plate weirs for the measurement of flow of clear water in open channels under free flow conditions. It includes the requirements for the use of full-width rectangular thin-plate weirs in submerged (drowned) flow conditions.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 772, *Hydrometry — Vocabulary and symbols*

3 Terms and definitions

For the purposes of this document, the terms and definitions given in ISO 772 apply.

4 Symbols and abbreviated terms

A	m^2	area of approach channel
B	m	width of approach channel
b	m	measured width of the notch
b_{max}	m	width of notch at maximum head (V-notch)
C		discharge coefficient (gauged head)
C_d		coefficient of discharge
f		drowned flow reduction factor
fC		combined coefficient of discharge
C_v		coefficient of velocity
e_b	m	random uncertainty in the width measurement
g	m/s^2	acceleration due to gravity
H	m	total head above crest level
h	m	upstream gauged head above crest level (upstream head is inferred if no subscript is used)
J		numerical constant
l	m	distance of the head measurement section upstream of the weir
n		number of measurements in a set
p	m	height of the crest relative to the floor
Q	m^3/s	volumetric rate of flow
S		submergence ratio, h_2/h_1